



***Earn**₹s*

DSA in C++

Loops

Lecture 09

Summary Notes

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Predict the output:

```
int main() {  
  ✓ int i;  
  ✓ while (i = 10) {  
    cout << i << endl;  
    i = i + 1;  
  }  
}
```

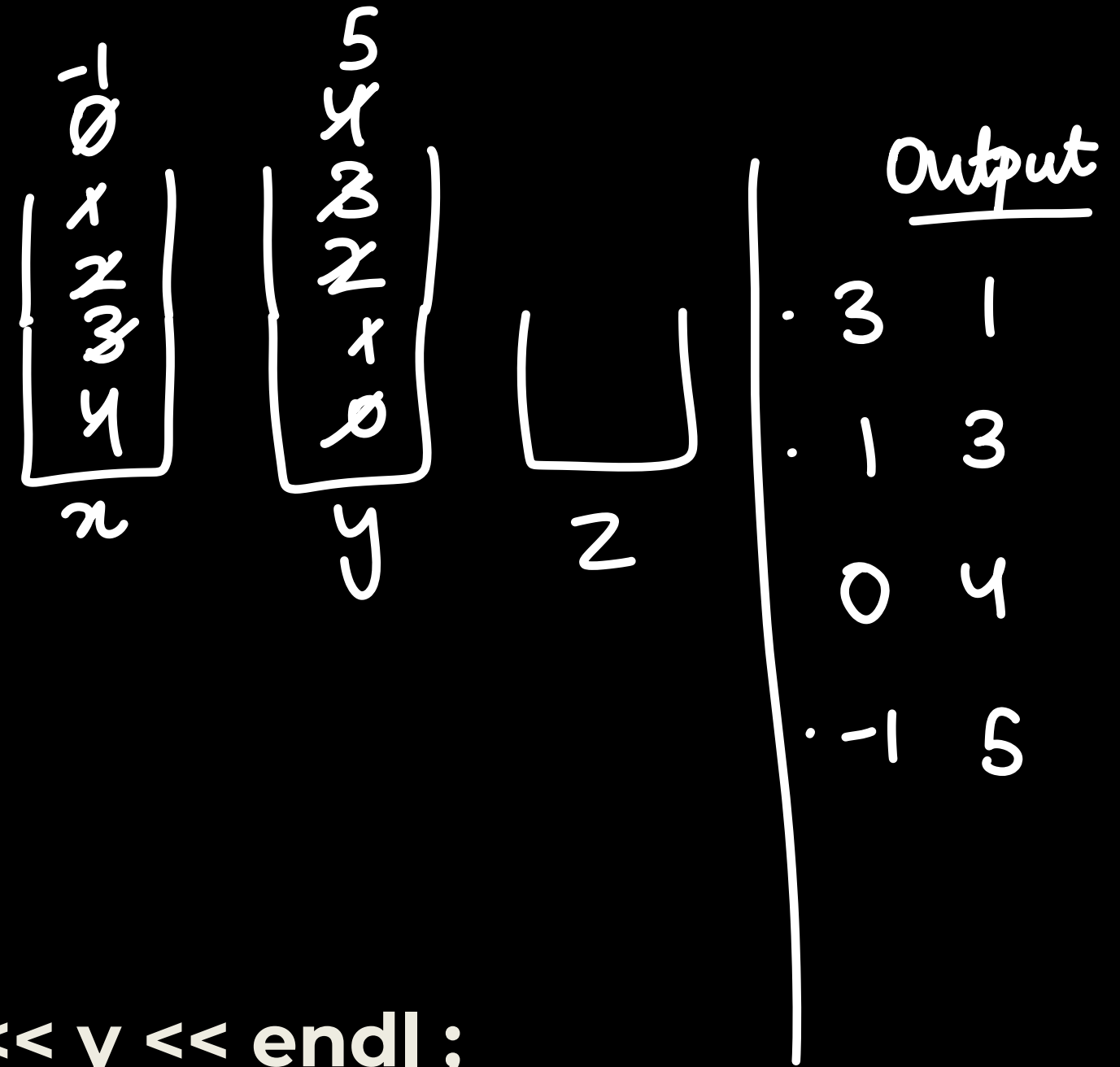
//
10
11
10

10
10

infinite

Predict the output:

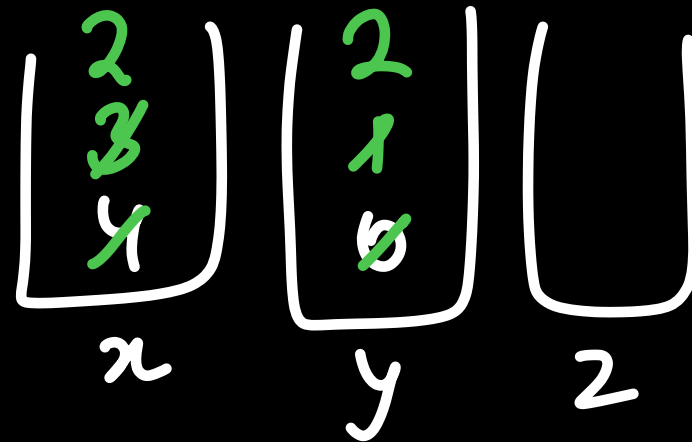
```
int main() {  
    int x = 4, y = 0, z;  
    while ( x >= 0 ) {  
        x-- ;  
        y++ ;  
        if ( x == y )  
            continue ;  
        else  
            cout << x << " " << y << endl ;  
    }  
}
```



Predict the output:

```
int main() {
    ✓int x = 4, y = 0, z;
    while ( x >= 0 ) {
        |   if ( x == y )
        |       break;
        |   else
        |       cout << x << " " << y << endl;
        |       x--;
        |       y++;
        |   }
}
```

Diagram illustrating the initial state of variables x, y, and z. Box x contains 4, box y contains 0, and box z is empty. All values are marked with a green checkmark.



Output

4	0
3	1

Predict the output: HW

```
int main() {  
    ✓ int t = 10;  
    ✓ while(t/=2){  
        cout << "Hello" << endl;  
    }  
}
```

0
1
2
8
10
t

Output

- Hello
- Hello
- Hello
-

t-- means , pehle isternal karo , fir ghatao
Predict the output:

```
✓ int t = 4;  
  while(t--){  
      cout<<"Hello"<<endl;  
  }
```

1
0
1
2
3
4
t

Output

- Hello
- Hello
- Hello
- Hello

0 → false

Loop exit

Ques: Count digits of a number

take 'n' input

$n = 13219$ ~~1321~~ ~~132~~ ~~13~~ ~~1~~ ~~0~~

Count = 0 1 2 3 4 5

Apurva



while($n \neq 10$)

```
int count = 0;
while (n > 0) {
    n /= 10,
    count ++,
}
cout << count;
```

~~✖✖~~ $n \neq 10$, it loses a digit

Ques: Print sum of digits of a number

$$n = 15243$$

$$n = 9871 \quad \xrightarrow{n \% 10}$$

last digit

ones place

$$1+5+2+4+3 = 3+4+2+5+1$$

n	Sum	$n \% 10$
15243	0	3
1524	3	4
152	7	2
15	9	5
1	14	1
0	15	

Ques: Reverse of a number

$$n = 2973 \quad \text{rev} = 3792$$

$$0 \rightarrow 3 \rightarrow 30 \rightarrow 37 \rightarrow 370 \rightarrow 379 \rightarrow 3790 \rightarrow 3792$$

$$n = \cancel{2973} \quad \cancel{297} \quad \cancel{29} \quad \cancel{2} \quad 0$$

$$r = \cancel{00830} \quad \cancel{37} \quad \cancel{370} \quad \cancel{379} \quad \cancel{3790} \quad 3792$$

Q. Print the sum of a given no > its reverse.

Ques: Factorial of a number (whole nos)

$n!$ or $\underline{n}$

$$n! = n \times n-1 \times n-2 \dots 3 \times 2 \times 1$$

$$= 1 \times 2 \times 3 \dots n$$

$$\underline{5} = 5 \times 4 \times 3 \times 2 \times 1$$

$$= 120$$

Ques: 'a' raise to the power 'b'

($a \geq 0, b \geq 0$)

$$a^b = \underbrace{a \times a \times a \cdots}_{b\text{-times}}$$